

Efficiency Comparison of BB84 and B92 Quantum Key Distribution Protocols

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1 Introduction

Quantum Key Distribution (QKD) enables two distant parties, traditionally called Alice and Bob, to establish a shared secret key whose security is guaranteed by the laws of quantum mechanics. Among the earliest and most studied QKD protocols are the BB84 protocol and the B92 protocol.

In this work, we compare BB84 and B92 in terms of their *efficiency*, defined as the amount of secret key generated per qubit sent through the quantum channel. We focus on the idealized, noiseless case in order to isolate the intrinsic efficiency of each protocol.

2 Definition of Efficiency

We define the efficiency η of a QKD protocol as

$$\eta = \frac{\text{number of secret key bits}}{\text{number of qubits sent}}. \quad (1)$$

This definition accounts for:

- basis mismatches,
- inconclusive measurement outcomes,
- public discussion and sifting.

Error correction and privacy amplification are not considered explicitly, as they affect both protocols similarly in the ideal case.

3 The BB84 Protocol

3.1 Protocol Description

In the BB84 protocol, Alice prepares qubits randomly in one of four states:

$$|0\rangle, |1\rangle \quad (\text{Z basis}), \quad |+\rangle, |-\rangle \quad (\text{X basis}),$$

where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

Alice randomly chooses one of the two bases (Z or X) for each qubit. Bob measures each incoming qubit in a randomly chosen basis (Z or X).

3.2 Sifting Process

After transmission, Alice and Bob publicly compare their basis choices and discard all events in which different bases were used.

Since Alice and Bob choose their bases independently and uniformly at random, the probability that they use the same basis is

$$P_{\text{same}} = \frac{1}{2}. \quad (2)$$

Thus, on average, half of the transmitted qubits contribute one raw key bit.

3.3 Efficiency of BB84

In the ideal, noiseless case:

- Each transmitted qubit yields a key bit with probability $1/2$.
- No inconclusive measurements occur.

Therefore, the efficiency of BB84 is

$$\eta_{\text{BB84}} = \frac{1}{2}. \quad (3)$$

That is, Alice and Bob obtain one secret key bit for every two qubits sent.

4 The B92 Protocol

4.1 Protocol Description

The B92 protocol uses only two non-orthogonal quantum states. Alice encodes bits using the states

$$|\psi_0\rangle, \quad |\psi_1\rangle,$$

where

$$|\langle\psi_0|\psi_1\rangle| \neq 0.$$

A common choice is

$$|\psi_0\rangle = |0\rangle, \quad |\psi_1\rangle = |+\rangle.$$

Bob performs a measurement designed to unambiguously discriminate between the two states. Such a measurement has three possible outcomes:

- a conclusive identification of $|\psi_0\rangle$,
- a conclusive identification of $|\psi_1\rangle$,
- an inconclusive result.

4.2 Conclusive Measurement Probability

For two non-orthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$, the maximum probability of obtaining a conclusive outcome is

$$P_{\text{conclusive}} = 1 - |\langle\psi_0|\psi_1\rangle|. \quad (4)$$

For the choice $|0\rangle$ and $|+\rangle$,

$$|\langle 0|+\rangle| = \frac{1}{\sqrt{2}},$$

so

$$P_{\text{conclusive}} = 1 - \frac{1}{\sqrt{2}} \approx 0.293. \quad (5)$$

Thus, fewer than one-third of the transmitted qubits yield a usable raw key bit.

4.3 Efficiency of B92

In the ideal case:

- Only conclusive outcomes contribute to the key.
- Inconclusive outcomes are discarded.

Therefore, the efficiency of B92 is

$$\eta_{\text{B92}} = 1 - |\langle\psi_0|\psi_1\rangle|. \quad (6)$$

For the standard implementation,

$$\eta_{\text{B92}} \approx 0.293. \quad (7)$$

5 Comparison of BB84 and B92

We summarize the efficiencies:

Protocol	Efficiency (bits per qubit)
BB84	0.5
B92	≈ 0.293

Clearly,

$$\eta_{\text{BB84}} > \eta_{\text{B92}}. \quad (8)$$

6 Conclusion

In terms of secret key generated per qubit sent through the quantum channel, the BB84 protocol is more efficient than the B92 protocol. This difference arises from the fact that BB84 employs orthogonal states within each basis, leading to deterministic measurement outcomes when the bases match, whereas B92 relies on non-orthogonal state discrimination, which necessarily produces inconclusive results.

BB84 is more efficient than B92 in terms of secret key per transmitted qubit.

While B92 has conceptual simplicity and requires fewer quantum states, this advantage comes at the cost of reduced key generation efficiency.